

Logical address and Physical address

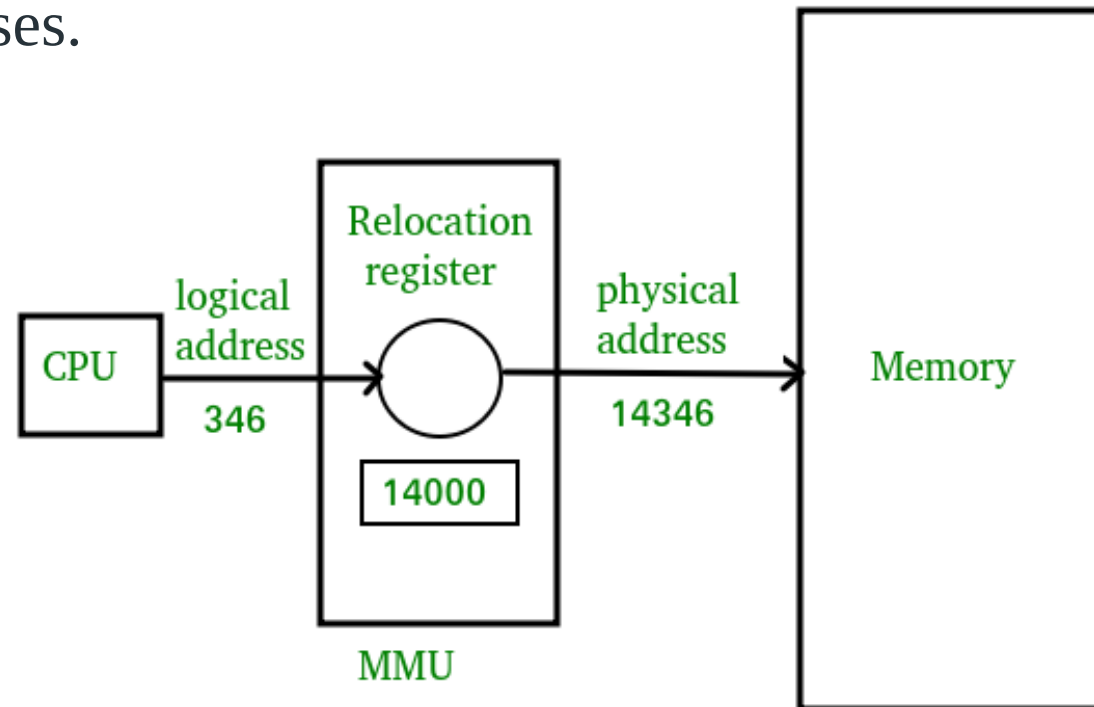
Prepared by–
Bipasha Majumder
Lecturer, Department of CSE, DIU

Logical Address

A logical address, also known as a virtual address, is an address generated by the CPU during program execution.

Physical Address

A physical address is the actual address in the main memory where data is stored. It is a location in physical memory, as opposed to a virtual address. Physical addresses are used by the Memory Management Unit (MMU) to translate logical addresses into physical addresses.



Parameter	LOGICAL ADDRESS	PHYSICAL ADDRESS
Basic	generated by CPU	location in a memory unit
Address Space	Logical Address Space is set of all logical addresses generated by CPU in reference to a program.	Physical Address is set of all physical addresses mapped to the corresponding logical addresses.
Visibility	User can view the logical address of a program.	User can never view physical address of program.
Generation	generated by the CPU	Computed by MMU
Access	The user can use the logical address to access the physical address.	The user can indirectly access physical address but not directly.
Editable	Logical address can be change.	Physical address will not change.
Also called	virtual address.	real address.

Memory size of 8086

- The **Intel 8086** microprocessor has a **20-bit address bus**.
- An address bus determines the range of memory addresses the processor can access.

With 20 address lines, the total number of unique memory locations that can be addressed is:

$$\begin{aligned}2^{20} &= 1,048,576 \text{ bytes} \\ &= 1 \text{ Megabyte (MB) bytes} \\ &= 1 \text{ Megabyte (MB)}\end{aligned}$$